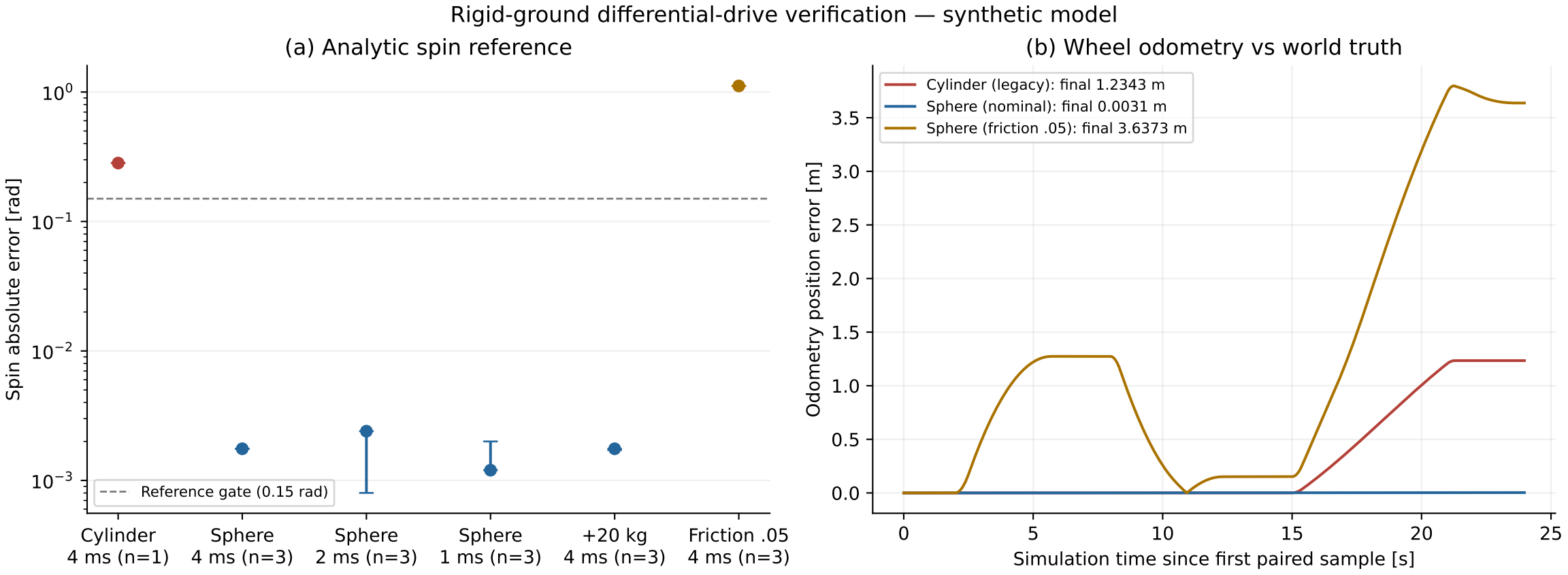
Rigid-ground differential-drive verification — synthetic model
(a) Analytic spin reference
Spin absolute error [rad]
10^{0}
10^{-1}
10^{-2}
10^{-3}
Reference gate (0.15 rad)
Cylinder
4 ms (n=1)
Sphere
4 ms (n=3)
Sphere
2 ms (n=3)
Sphere
1 ms (n=3)
+20 kg
4 ms (n=3)
Friction .05
4 ms (n=3)
(b) Wheel odometry vs world truth
Cylinder (legacy): final 1.2343 m
Sphere (nominal): final 0.0031 m
Sphere (friction .05): final 3.6373 m
Odometry position error [m]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0
5
10
15
20
25
Simulation time since first paired sample [s]